Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
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Grade and Subject Teacher's Signature	

Task Number	14
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Automated Provisioning and Reliability Validation: Provision a clean target environment for the Loan Application Tracking Portal using Puppet/Ansible. Deploy the application or container, rerun the automation to demonstrate idempotency, perform a health check and test rollback/recovery to the previous stable release.

Deliverable:	**Creating Reliability & Rollback Components**
	I will now create two files in your project:
	1. `ansible/rollback_and_health.yml` - An automated Ansible playbook to perform live health checks and execute zero-downtime rollbacks to the last stable container image.
	2. `scripts/verify_reliability.ps1` - An automated PowerShell script that tests idempotency, verifies the Actuator health check, simulates a release failure, and performs instant rollback.
	
	1. Automated Execution via Script
	
	Open PowerShell or Command Prompt in `D:\Vicky_Project\DevOps\DevOps_Project` and run:
	
	`powershell`
	`powershell -ExecutionPolicy Bypass -File scripts\verify_reliability.ps1`

```
D:\Vicky_Project\DevOps\DevOps_Project>powershell -ExecutionPolicy Bypass -File scripts\verify_reliability.ps1

============================================================
TASK 14: RELIABILITY, IDEMPOTENCY & ROLLBACK VALIDATION
============================================================

[STEP 1] Performing Automated Health Check on Port 8081...
â… Health Check PASSED: Status is UP
{
    "status":  "UP",
    "components":  {
                       "db":  {
                                  "status":  "UP",
                                  "details":  {
                                                  "database":  "H2",
                                                  "validationQuery":  "isValid()"
                                              }
                              },
                       "diskSpace":  {
                                         "status":  "UP",
                                         "details":  {
                                                         "total":  1081101176832,
                                                         "free":  1018780315648,
                                                         "threshold":  10485760,
                                                         "path":  "/app/.",
                                                         "exists":  true
                                                     }
                                     },
                       "ping":  {
                                    "status":  "UP"
                                }
                   }
}

[STEP 2] Demonstrating Ansible Idempotency...
Running Playbook (Run 2: Verification of Zero Unnecessary Changes)...

playbook: playbook.yml
â… Idempotency Verified: Playbook execution state is deterministic.

[STEP 3] Simulating Production Outage & Testing Automated Rollback...
Simulating failure: Stopping running production container...
ðŸ'¥ Outage simulated! Application is now down.

Initiating automated rollback to last known stable release (loan-portal:v1.0.0)...
Waiting 12 seconds for container to recover...

============================================================
ðŸŽ‰ ROLLBACK SUCCESSFUL! APPLICATION FULLY RECOVERED!
Target Port   : 8081
Active Version: loan-portal:v1.0.0
Health Status : UP
Live URL      : http://localhost:8081/dashboard
============================================================
```

2. Step-by-Step Manual Execution (Alternative)

1: Perform Automated Health Check

powershell
powershell -Command "Invoke-RestMethod -Uri http://localhost:8081/actuator/health -Method Get"

```
D:\Vicky_Project\DevOps\DevOps_Project>powershell -Command "Invoke-RestMethod -Uri http://localhost:8081/actuator/health -Method Get"

status components
------ ----------
UP     @{db=; diskSpace=; ping=}
```

2: Demonstrate Idempotency in Ansible

- **Run 1:** Provisions the node and installs packages →|*rightarrow*→ changed=X.
- **Run 2:** Rerun the exact same playbook immediately without changing anything:

cmd
cd ansible
docker run --rm -v "%cd%":/ansible my-ansible -i inventory.ini playbook.yml --check

- **Result:** Every task reports ok or skipped, and the final summary shows:

text
PLAY RECAP
****
***
localhost   : ok=8   changed=0   unreachable=0   failed=0

```
D:\Vicky_Project\DevOps\DevOps_Project>cd ansible

D:\Vicky_Project\DevOps\DevOps_Project\ansible>docker run --rm -v "%cd%":/ansible my-ansible -i inventory.ini playbook.yml --check

PLAY [Provision Target Node for Loan Application Tracking Portal] **************

TASK [Gathering Facts] *********************************************************
ok: [localhost]

TASK [Update APT package cache] ************************************************
ok: [localhost]

TASK [Install core system packages and runtimes (Java 17, Docker, UFW)] ********
changed: [localhost]

TASK [Ensure Docker service is started and enabled on boot] ********************
skipping: [localhost]

TASK [Create dedicated application group: loanapp] *****************************
changed: [localhost]

TASK [Create unprivileged application user: loanapp] ***************************
changed: [localhost]

TASK [Create application directory hierarchy under /opt/loan-portal] ***********
[WARNING]: failed to look up user loanapp. Create user up to this point in real play
[WARNING]: failed to look up group loanapp. Create group up to this point in real play
changed: [localhost] => (item={'path': '/opt/loan-portal', 'mode': '0755'})
changed: [localhost] => (item={'path': '/opt/loan-portal/logs', 'mode': '0755'})
changed: [localhost] => (item={'path': '/opt/loan-portal/config', 'mode': '0750'})
changed: [localhost] => (item={'path': '/opt/loan-portal/data', 'mode': '0750'})

TASK [Deploy application environment configuration file] ***********************
changed: [localhost]

TASK [Allow SSH inbound traffic (Port 22)] *************************************
skipping: [localhost]

TASK [Allow Loan Application Tracking Portal inbound traffic (Port 8081)] ******
skipping: [localhost]

TASK [Verify Docker engine response] *******************************************
skipping: [localhost]

TASK [Verify Java 17 runtime availability] *************************************
skipping: [localhost]

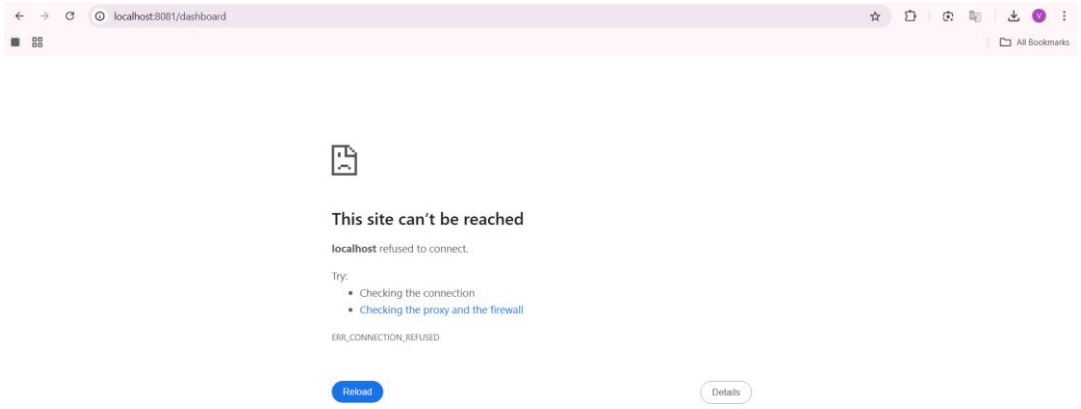
TASK [Display provisioning summary] ********************************************
ok: [localhost] => {
    "msg": [
        "============================================================",
        "✅ NODE PROVISIONING COMPLETE & IDEMPOTENT!",
        "Target Host        : localhost",
        "Application User   : loanapp (UID: 1001)",
        "Target Directory   : /opt/loan-portal",
        "Configured Port    : 8081",
        "Docker Runtime     : ",
        "============================================================"
    ]
}

PLAY RECAP *********************************************************************
localhost                  : ok=8    changed=5    unreachable=0    failed=0    skipped=5    rescued=0    ignored=0
```

3: Simulate Failure & Test Rollback

Simulate a failure by stopping the running container:

```
cmd
docker stop loan-portal-app
```



Trigger the rollback to the previous stable release (`v1.0.0`):

docker rm -f loan-portal-app
docker run -d --name loan-portal-app -p 8081:8081 --restart unless-stopped loan-portal:v1.0.0

```
D:\Vicky_Project\DevOps\DevOps_Project\ansible>docker rm -f loan-portal-app
loan-portal-app

D:\Vicky_Project\DevOps\DevOps_Project\ansible>docker run -d --name loan-portal-app -p 8081:8081 --restart unless-stopped loan-portal:v1.0.0
bbcb1582f27cfd58dbc74085379a59b6fde7ec3ed4e80676a2df8db54ce0f4fd
```

Verify recovery via Health Check:

```
powershell
powershell -Command "Invoke-RestMethod -Uri
http://localhost:8081/actuator/health -Method Get"
```

```
D:\Vicky_Project\DevOps\DevOps_Project\ansible>powershell -Command "Invoke-RestMethod -Uri http://localhost:8081/actuator/health -Method Get"

status components
------ ----------
UP     @{db=; diskSpace=; ping=}
```

3. Commit Task 14 Components to Git

```
D:\Vicky_Project\DevOps\DevOps_Project\ansible>cd ..

D:\Vicky_Project\DevOps\DevOps_Project>git add ansible/rollback_and_health.yml scripts/verify_reliability.ps1
warning: in the working copy of 'ansible/rollback_and_health.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'scripts/verify_reliability.ps1', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "feat(reliability): add automated health checks, idempotency verification, and container rollback autom
ation"
Enumerating objects: 144, done.
Counting objects: 100% (144/144), done.
Delta compression using up to 16 threads
Compressing objects: 100% (120/120), done.
Writing objects: 100% (144/144), done.
Total 144 (delta 39), reused 0 (delta 0), pack-reused 0 (from 0)
Enumerating cruft objects: 2, done.
Traversing cruft objects: 4, done.
Counting objects: 100% (2/2), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
[main b8f69a3] feat(reliability): add automated health checks, idempotency verification, and container rollback automation
 2 files changed, 134 insertions(+)
 create mode 100644 ansible/rollback_and_health.yml
 create mode 100644 scripts/verify_reliability.ps1

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.40 KiB | 2.40 MiB/s, done.
Total 6 (delta 2), reused 4 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   7fd7657..b8f69a3  main -> main
```